

“数字集成电路设计” 2023-2024 第 1 学期 第 5 周作业

参考答案:

6.10 参考答案

a: P 管上升时间:

 $V_{dsat,p} = 0.51$, $V_{out} \leq 0.69$ 时 P 管饱和区, $V_{out} > 0.69$ 时线性。

$$I_{avg, rise} = \frac{1}{2} [I(V_{in} = 0V, V_{out} = 0.12V) + I(V_{in} = 0V, V_{out} = 1.08V)]$$

$$= 179.0 \mu A$$

$$\tau_{rise} = \frac{C_{load} \cdot \Delta V}{I_{avg, rise}} = 53.6 ps$$

N 管下降时间:

 $V_{dsat,n} = 0.25$, V_{out} 大于 0.25, N 管饱和, V_{out} 小于 0.25 时 N 管线性。

$$I_{avg, fall} = \frac{1}{2} [I(V_{in} = 1.2V, V_{out} = 1.08V) + I(V_{in} = 1.2V, V_{out} = 0.12V)]$$

$$= 145.2 \mu A$$

$$\tau_{fall} = 66.1 ps$$

b:

$$f_{max} = \frac{1}{\max(t_{fall}, t_{rise}) \times 2} = 7.6 \times 10^9$$

c:

$$P_{avg, max} = C_{load} V_{DD}^2 f_{max} = 109.4 \mu W$$

d:

$$\left(\frac{W}{L}\right)_p = 40 \qquad \left(\frac{W}{L}\right)_n = 26.7 \qquad V_{TH} \text{ 不变}$$

第 6 章补充习题

1、答:

$$1) C_g = C_i = C_d = 10 fF, C_L = 20 pF$$

$$\tau_{p0} = 70 ps, \quad N+1=3$$

$$\alpha^{N+1} = \alpha^3 = \frac{C_{load}}{C_g} = \frac{20 \times 10^{-12}}{10 \times 10^{-15}}$$

$$\therefore \alpha = 12.6$$

$$\tau \approx 2.856ns$$

尺寸分别为 12.6 和 158.76

$$2) \gamma=1, \text{则 } \alpha_{opt} = 3.6, N + 1 = \frac{\ln 2000}{\ln 3.6} \approx 5.9339$$

$$\text{取 } N=6, \tau = 7\tau_{p0} \left(1 + \frac{\alpha C_g}{C_d}\right) \approx 1.94ns$$

$$\text{当 } N=5 \text{ 时, } \tau = 6\tau_{p0} \left(1 + \frac{\alpha C_g}{C_d}\right) \approx 1.91ns$$

$$\text{当 } N=4 \text{ 时, } \tau = 5\tau_{p0} \left(1 + \frac{\alpha C_g}{C_d}\right) \approx 1.95ns$$

因此, $N+1=6, \tau = 1.91ns$

3) 方法一较方法二而言, 延时相对较高, 但是占用面积较小, 功耗更低。

2、答:

$$\begin{aligned} \tau_{AtoB} &= R_1 C_1 + R_1 C_2 + R_1 C_4 + (R_1 + R_3) C_3 + (R_1 + R_3 + R_5) C_5 \\ &= 27500fs = 27.5ps \end{aligned}$$

$$T_p = 0.69\tau_{AtoB} = 18.975ps$$